

Low-Precision Hardware Architectures Meet Recommendation Model Inference at Scale

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Abstract—Tremendous success of machine learning (ML) and the unabated growth in ML model complexity motivated many ML-specific designs in both CPU and accelerator architectures to speed up the model inference. While these architectures are diverse, highly optimized low-precision arithmetic is a component shared by most. Impressive compute throughputs are indeed often exhibited by these architectures on benchmark ML models. Nevertheless, production models such as recommendation systems important to Facebook’s personalization services are demanding and complex: These systems must serve billions of users per month responsively with low latency while maintaining high prediction accuracy, notwithstanding computations with many tens of billions parameters per inference. Do these low-precision architectures work well with our production recommendation systems? They do. But not without significant effort. We share in this paper our search strategies to adapt reference recommendation models to low-precision hardware, our optimization of low-precision compute kernels, and the design and development of tool chain so as to maintain our models’ accuracy throughout their lifespan during which topic trends and users’ interests inevitably evolve. Practicing these low-precision technologies helped us save datacenter capacities while deploying models with up to 5X complexity that would otherwise not be deployed on traditional general-purpose CPUs. We believe these lessons from the trenches promote better co-design between hardware architecture and software engineering and advance the state of the art of ML in industry.

I. INTRODUCTION

Machine learning has experienced revolutionary advances in the past decade. The advancement has led to state-of-the-art production models that are complex, compute intensive and consume large amount of memory resources [17, 27, 40, 60, 65, 66]. For instance, the memory requirement of deep learning recommender systems has grown from few tens of GBs to the terabyte regime [41, 63]. To fuel the needs of rising application domains, computer architectures are evolving, be it general-purpose CPUs or GPUs with new features [1, 21] or specialized accelerators [2, 12, 34], targeting deep learning. While these architectural features or architectures are diverse, one major commonality is the support for low-precision arithmetic datatype.

Low-precision arithmetic datatype offers several orthogonal advantages simultaneously. Storing model parameters in low-precision formats greatly increases parameter loading throughput and reduces power consumed on data movement [16, 25,

26, 35, 52]. It also increases compute throughput if computation is well optimized for the underlying architecture. For instance, TPUs are customized [34] with the int8 and fp16 arithmetic to accelerate the inference of neural networks. The latest Intel CPUs added the VNNI feature (Vector Neural Network Instructions) [9] to support neural network acceleration. NVIDIA Tensor Cores support mixed precision of fp32 and fp16 to deliver 7x more flops/s than fp32 on a Tesla V100 GPU [42]. Moreover, int8 is generally supported on many customized accelerators and CPUs, which usually deliver much higher compute throughput than floating point.

To improve user experience, social network services such as those Facebook provides, leverage deep learning in a wide range of applications [27]. Among all the deep learning applications, recent work showed, in 2019, more than 70% of the total machine learning inference cycles served by Facebook’s datacenter fleets are devoted to recommendation use cases [24]. Thus, performance and energy efficiency improvement of deep learning recommendation models will translate into significant infrastructure capacity saving.

Recommender systems have evolved with increasing model complexities: from simple content and collaborative filtering, to matrix factorization [37, 50], multi-layer perceptrons and deep learning models [18, 47, 54, 55, 64, 65, 67]. Facebook’s production recommendation models consist of many tens to a hundred billion parameters to achieve high learning capacity [45]. Despite the continuously growing learning capacity, the inference latency of the recommender systems must meet specified SLA thresholds, such as 100ms, to ensure quality user experience [48]. Low-precision arithmetic is promising to improve the inference efficiency. Nevertheless, deploying low precision in production recommender models at Facebook’s scale proves extremely challenging:

Meeting Stringent Accuracy Requirement: Facebook’s recommendation models are typically trained using the full-precision numerical format. Trained models can be converted to a lower precision format for deployment through post-training quantization (PTQ). We do not deploy a low-precision recommendation model that has more than 0.05% accuracy degradation compared against the original full-precision model [58]. This is a much more stringent requirement compared to other vision or language use cases, not to mention that